PROJECT NOBLE · HANDOFF BRIEF

Halo: Reach Community Edition

Project Noble — the standalone front-end client

Prepared 30 October 2026, 16:25 ET · for the next engineer or agent picking this up
Everything below is verified against the current build and the live repository.

Repository	D:\Halo-Reach-Community-Edition (remote: github.com/O7OF/test-build-01, branch main)
Version	0.8.3 (Menu Cleanup & Controls) — src/common/Version.hpp
Build	CMake Release, MSVC 2022 — 0 errors / 0 warnings expected
Owner	O7OF hosts the repo and server. Tyler (WeabooTyler117) plays and directs; he is not a C++ developer.

1. What this project is

Project Noble runs Halo: Reach from the Master Chief Collection install as a standalone game with its own front-end. The client loads haloreach.dll without MCC's launcher and draws its own interface — main menu, campaign, multiplayer and Firefight setup, server browser, settings, career and Armory — as an ImGui overlay over the engine. MCC is treated as read-only data; Steam ownership is checked.

The point of the project is that Reach's original menus are clunky and hard to read on modern displays. Noble replaces them with a cleaner, denser, controller-friendly interface while leaving the game itself untouched.

2. Build and verify

Stop the client first, then build. A running client produces LNK1104.

```
taskkill /IM noble-client.exe /F
"C:\Program Files\Microsoft Visual Studio\2022\Community\Common7\IDE\CommonExtensions\
 Microsoft\CMake\CMake\bin\cmake.exe" --build build --config Release -- -m
```

Do not use build.bat. UI work is verified with screenshots, not reasoning: the capture harness renders one screen, writes a BMP and exits, so each change can be inspected as an image.

```
noble-client.exe -menu-screen armory -menu-capture 90
```

- **Screens:** main, campaign, skulls, firefight, custom, forge, servers, settings, armory, mods, career, lobby, social.
- **-menu-tab N:** also selects the Settings or Career tab, and VANILLA (0) / MODDED (1) on the setup screens.
- **Other hooks:** -test-map-gametypes (headless suites, writes gametype-selftest.log, exit 0/1), -auto-play, -auto-play-variant <stem>, -auto-play-map <scenario>, -pause-at <s>, -scoreboard-at <s>, -quit-after <s>, -armory-scene-index <n>, -armory-focus "<slot>".
- **Scripts:** tools\test-modes.ps1 drives real launches (15 game types; -Maps adds Slayer on all 20 maps). tools\check-settings-wired.ps1 audits every setting for a real consumer.
- **Important:** a bare command-line launch cannot start a match. It needs -player-name, -service-tag, -account-user, -account-token, -user-id and -account-dev, which the menu hands to the relaunched process.

3. What changed in the most recent session

Ten commits, all pushed and clean. The Armory thread ran through several approaches; the ones marked superseded are recorded so nobody repeats them.

Commit	What it changed
`6048d9a`	Career: bonusCredits — granted cR that is spendable but never counts toward rank.
`da95dcb`	Armory: deleted the OBJ preview entirely (renderer, layer builder, setting and row).
`c3a93ec`	Armory: back on the in-game model; the OBJ re-export is a different mesh and was rejected.
`cafb0f4`	Docs: what MCC actually ships for emblems, and the two possible routes.
`d89944e`	Superseded — OBJ renderer: eye-strafe pan, made the default for one revision.
`d0026f1`	Superseded — designed stage behind the OBJ renderer.
`a159559`	Armory cameras mapped by capture; test hooks added; left/right shoulder proven impossible.
`80eba4a`	Superseded — armoryPreview switch and renderer evaluation.
`d136a00`	Relaunch keeps the old window on screen until the new process has one: no desktop flash.
`9308d71`	Armory: appearance writes coalesced to 250 ms, so holding an arrow key stops the model reload.

4. Player save state — handle with care

Tyler asked for a career reset that keeps money: level back to Recruit, with 1,000,000 cR available so he can walk the unlock path again.

career.json	credits (lifetime earned) 0 · creditsSpent 0 · bonusCredits 1,000,000 · armoryOwned empty · armoryUnlockAll false · all stats, history, challenges and commendations cleared.
Backup 1	data\userdata\career.backup-20261030-154234.json — the entire pre-reset career (rank 49, 1e9 cR).
Backup 2	data\appearance.before-reset-20261030.json — the original Spartan: Hilands helmet, Emile chest plate, Kat tricorder, Linda shoulder, Adrianna knee, blue flames, Jorge voice, colours 15/14.

The trap that bit us: clearing armoryOwned makes the Armory treat every equipped piece as locked, revert it to DEFAULT, and then save that over data\appearance.json. Never clear ownership without backing that file up first.

5. Open work, in priority order

5.1 Emblem background and foreground picker (in progress)

Tyler wants Reach's emblem editor: choose a background shape and a foreground symbol separately. The data reality first, because it shapes the design:

- **The data:** Data\UI\playercustomization.xml in the MCC install defines 330 emblems as image1="e_002_p" image2="e_002_s". Those are the primary and secondary colour layers of one finished emblem — not a reusable backdrop plus symbol pair — and the backdrops are effectively unique per emblem. Reach 360's own emblem sets are not in MCC or in Noble's data.
- **Already landed:** UiArt::GetEmblemBackdropLayers(), GetEmblemSymbolLayers() and EmblemLayerTexture() expose roughly 225 layers each, loaded by name through EmblemByName.
- **Remaining:** two Armory identity rows (the IDENTITY block in ArmorCustomizer.cpp, around line 1638), storage for the chosen pair, and composition at the two places emblems are drawn — FrontEndMenu.cpp:861 in

PlayerRow and FrontEndMenu.cpp:3660 in the Armory nameplate. Draw the backdrop layer, then the symbol layer.

- **Decision made:** Tyler approved prototyping the matrix from MCC's layers (any backdrop with any symbol) rather than waiting for the original Reach assets to be extracted.

5.2 Armory: the black chamber, and left versus right shoulder

- **Problem:** the Armory shows Reach's in-game scene, with the correct model in front of an unlit black chamber. A renderer swap was attempted and rejected because the replacement mesh was wrong. Code cannot composite behind the engine's 3D output, and five approaches proved the void is engine-drawn geometry.
- **Remaining route:** a data fix — light or replace the chamber in data\menumaps\mainmenu_all.map. ReachArmory already walks the GLOBALS table, which is the entry point for that work.
- **Working:** per-slot camera panning already works through the engine's own scenes, verified by capture: 0x0D helmet, 0x0F chest, 0x10 wrist, 0x11 utility, 0x12 knees, 0x13 shoulders, 0x14 colours, 0x0A and 0x0C full body. Indices from 0x15 up leave the Armory. 0x0E is the roster scene and is excluded.
- **Not possible yet:** the map has one shoulder camera, and rotating the model changes nothing because the camera rides with the character (measured 5 % pixel difference, all of it animated particles). It needs a camera with a lateral offset; the globals table only carries the scene index at base + 0x24 * stride.

5.3 Scoreboard modernisation

Reach's scoreboard is engine-drawn and still reads as the original game's. Either restyle the engine screen, for which no handle has been found, or draw a Noble overlay over it. The overlay needs per-player rows — name, team, score, kills and deaths — and the same data is already parsed for the post-game results screen, so this is data plumbing rather than new reverse engineering. -scoreboard-at <s> exists for captures.

5.4 Campaign CONTINUE should resume the last checkpoint

CONTINUE currently relaunches the last campaign mission from its start; the row and the launch choke point are in FrontEndMenu.cpp. Real checkpoint resume needs Reach's save and checkpoint state, which nothing in Noble reads yet. The first step is locating the checkpoint block in the save data by comparing two saves taken at different points in the same mission, then passing it to the new process.

5.5 Truly seamless transitions

Every match or mission still crosses a process boundary: a new noble-client.exe starts and the old one exits. Commit d136a00 turned that into a window-for-window swap, but it is not seamless. A single-process transition needs the engine's own session control. BlamEngine::ExecuteCommand() dispatches through the engine vfunc[9] but nothing calls it and BlamEngine is not in the running client's path. The proposed experiment is a -exec-at <seconds> <command> hook, the same shape as -pause-at, dispatching candidates such as game_end, game_start, map_reload and quit mid-match to see whether the engine returns to the menu instead of tearing down. Earlier notes say the engine cannot switch maps once a match has started, so treat this as research.

5.6 Older backlog

Vote-to-kick needs a NetworkTransport kick message plus a roster in the pause menu and on TAB. CHANGE TEAM needs an engine action, and the row should only appear when a session reports teams. A real solo pause has no engine command yet. Engine-side graphics — texture filtering, shadows, motion blur, draw distance — can only be changed in MCC itself.

6. Traps and house rules

1. **Always update the handoff:** every change gets a new entry at the top of docs\HANDOFF.md headed "## 0. Pick up here (date, time ET) - what changed (name)", then mirror it to the Downloads folder as HANDOFF (2).md, PROJECT-NOBLE-CONTINUATION.md and MENU-REDESIGN-SPEC.md, commit, push, and close the client.
2. **Career semantics:** credits in career.json is lifetime earned; the spendable balance is credits - creditsSpent + bonusCredits, and rank is RankIndex(credits). Never hand out rank by raising credits; use bonusCredits.
3. **data\appearance.json is live data:** the client rewrites it on exit. Read section 4 before touching ownership.
4. **settings.json playerName:** has been clobbered to "Bob" during account-less capture runs. Check identity after any run that saves settings.
5. **Layout contract:** RightColumnLayout(ui) produces actions, strip and content; use HeroBannerHeight() and BlurbBlockHeight() (two-line cap) and PlayerRow(), and never pin rows to actionsY.
6. **SettingsPanel::Row() always returns true:** it is a layout helper. Action rows must test m_rowHovered && ImGui::IsMouseClicked(0), or focused plus Accept. Ignoring this once opened a file dialog every frame.
7. **No texture creation on the fallback draw path:** warm art in FrontEndMenu::Initialize through UiArt::WarmMenuBackground().
8. **Post-processing:** GammaPass in ReachHost.cpp is the only place Noble can change the picture. Its C++ GammaCB must match the HLSL cbuffer exactly — 16 versus 32 bytes crashed it — and it must never pass a null array to PSSetSamplers.
9. **Relaunching:** include the account flags from section 2, and forward test switches to the child process.
10. **Windows PowerShell 5.1:** no ternary operator; assigning a missing JSON key fails, so use Add-Member -Force; never assign to $args.
11. **A minimised window stops presenting:** so captures and -pause-at never fire. Keep the window visible while testing.
12. **Two Desktop folders exist:** the active one is C:\Users\tyler\Desktop, where the Project Noble shortcut lives.

7. How Tyler works

"Do whatever you recommend" and "crank it out" are standing instructions, and he often leaves the session running. Act autonomously, verify visually with captures, and report in plain language. Ask only when a genuine decision is needed — which of two data routes to take, or permission to destroy saved data. He will reject anything that looks like the wrong model or a placeholder, so compare against his screenshots before calling something finished.

Companion documents: docs\HANDOFF.md (detailed chronological log), docs\CONTINUATION.md (project map and commit table), docs\menu-redesign-spec.md (design spec).